O*NET OnLine

Network and Computer Systems Administrators

15-1244.00

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Install, configure, and maintain an organization's local area network (LAN), wide area network (WAN), data communications network, operating systems, and physical and virtual servers. Perform system monitoring and verify the integrity and availability of hardware, network, and server resources and systems. Review system and application logs and verify completion of scheduled jobs, including system backups. Analyze network and server resource consumption and control user access. Install and upgrade software and maintain software licenses. May assist in network modeling, analysis, planning, and coordination between network and data communications hardware and software.

Sample of reported job titles: LAN Administrator (Local Area Network Administrator), LAN Specialist (Local Area Network Specialist), Network Administrator (Network Admin), Network Analyst, Network Coordinator, Network Engineer, Network Manager, Network Specialist, Systems Engineer

Occupation-Specific Information

Tasks

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- Plan, coordinate, and implement network security measures to protect data, software, and hardware.
- Perform data backups and disaster recovery operations.
- Diagnose, troubleshoot, and resolve hardware, software, or other network and system problems, and replace defective components when necessary.
- Maintain and administer computer networks and related computing environments, including computer hardware, systems software, applications software, and all configurations.
- Configure, monitor, and maintain email applications or virus protection software.

Occupational Requirements

Work Activities

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- **Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.
- **Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
- **Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.
- **Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.
- **Updating and Using Relevant Knowledge** — Keeping up-to-date technically and applying new knowledge to your job.

Detailed Work Activities

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- Implement security measures for computer or information systems.
- Develop computer or information security policies or procedures.
- Create electronic data backup to prevent loss of information.
- Resolve computer network problems.
- Resolve computer software problems.

Work Context

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- **E-Mail** — 100% responded “Every day.”
- **Telephone Conversations** — 77% responded “Every day.”
- **Indoors, Environmentally Controlled** — 80% responded “Every day.”
- **Face-to-Face Discussions with Individuals and Within Teams** — 50% responded “Every day.”
- **Spend Time Sitting** — 50% responded “Continually or almost continually.”

Experience Requirements

Job Zone

Title	Job Zone Three: Medium Preparation Needed
Education	Most occupations in this zone require training in vocational schools, related on-the-job experience, or an associate's degree.
Related Experience	Previous work-related skill, knowledge, or experience is required for these occupations. For example, an electrician must have completed three or four years of apprenticeship or several years of vocational training, and often must have passed a licensing exam, in order to perform the job.
Job Training	Employees in these occupations usually need one or two years of training involving both on-the-job experience and informal training with experienced

	workers. A recognized apprenticeship program may be associated with these occupations.
Job Zone Examples	These occupations usually involve using communication and organizational skills to coordinate, supervise, manage, or train others to accomplish goals. Examples include hydroelectric production managers, desktop publishers, electricians, agricultural technicians, barbers, court reporters and simultaneous captioners, and medical assistants.
SVP Range	(6.0 to < 7.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

Apprenticeship Opportunities

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Example apprenticeship titles for this occupation:

- Information Assurance Specialist
- Information Technology Specialist
- It Systems Administrator
- Network and Computer Systems Administrator (Alternate Title: It Specialist)
- Network Support Technician

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Data base user interface and query software** — Amazon Elastic Compute Cloud EC2 ; Blackboard software; ServiceNow ; Transact-SQL

- **Development environment software** — Apache Kafka ; Apache Maven ; Go ; Microsoft PowerShell
- **Network monitoring software** — Nagios; Quest Foglight; Remote monitoring software; Wireshark
- **Operating system software** — Bash ; Microsoft Windows Server ; Red Hat Enterprise Linux ; UNIX Shell
- **Web platform development software** — Apache Tomcat ; Django ; Microsoft ASP.NET ; Node.js

Hot Technologies are requirements most frequently included across all employer job postings.
See all 95 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 18 In Demand skills for this occupation.

Essential Skills

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- **Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
- **Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
- **Monitoring** — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
- **Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
- **Speaking** — Talking to others to convey information effectively.

Transferable Skills

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- **Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
- **Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.
- **Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.
- **Troubleshooting** — Determining causes of operating errors and deciding what to do about it.
- **Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.

Knowledge

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- **Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
- **Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.
- **Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.
- **English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
- **Administration and Management** — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources modeling, leadership technique, production methods, and coordination of people and resources.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

30% Associate's degree or other 2-year degree required

27% Bachelor's degree required

20% Post-secondary certificate required

Worker Characteristics

Abilities

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- **Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
- **Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
- **Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).
- **Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.
- **Near Vision** — The ability to see details at close range (within a few feet of the observer).

Interests

All 3 displayed

Interest code: **CRI**

Want to discover your interests? Take the **O*NET Interest Profiler**.

- **Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.
  - **Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
- **Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
  - **Mechanics/Electronics** — Work involves maintaining and repairing of machinery, such as automotive engines, electronic equipment, computers, and communication systems.
  - **Engineering** — Work involves applying science and technology to the design, building, testing, and use of electrical and electronic components, mechanical devices and machines, automotive, marine, and aerospace equipment and vehicles, materials, or structures.
- **Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.

Work Styles

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- **Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
- **Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
- **Integrity** — A tendency to be honest and ethical at work.
- **Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.
- **Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025)	$47.66 hourly, $99,130 annual
State wages	Select a State Go
Local wages	ZIP Code Go
Employment (2025)	323,600 employees
Projected growth	Decline (-1% or lower)

(2025-2035)	
Projected job openings (2025-2035)	13,400
State trends	Select a State Go
Top industries (2025)	Professional, Scientific, and Technical Services Educational Services

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings	Select a State Go
Local job openings	ZIP Code Go

More Information

Related Occupations

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- 15-1241.00 Computer Network Architects Bright Outlook
- 15-1231.00 Computer Network Support Specialists
- 15-1211.00 Computer Systems Analysts
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1242.00 Database Administrators

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

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- American Library Association
- Association for Computing Machinery
- Healthcare Information and Management Systems Society
- IEEE Computer Society
- National Center for Women and Information Technology

Accreditation, Certification, & Unions

- CompTIA ALLY

- Building Industry Consulting Service International